

DA5000W Linear Algebra and Optimization

Assignment 1

October 2025

Topics

1. Solution for a system of linear equations
2. Rank-nullity theorem
3. linear dependency
4. vector space, basis, orthonormal basis
5. Linear map

Question 1: Find the Solution for the following system of linear equations

$$\begin{aligned}-2x + 3y + z &= 1 \\ -x + z &= 0 \\ 2y &= 5\end{aligned}$$

Question 2: Find the basis of the null space and the dimension

$$\begin{aligned}x + 5y + 2z \\ 2x + 4y + 3z \\ 4x + 8y + 6z\end{aligned}$$

Question 3: Decide whether or not the following vectors are linearly independent,

$$\mathbf{v1} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v2} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v3} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \quad \mathbf{v4} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Decide also if they span $\mathbb{R}^4$, by trying to solve $\mathbf{c1v1} + \mathbf{c2v2} + \mathbf{c3v3} + \mathbf{c4v4} = (0, 0, 0, 1)$.

Question 4: Let A be a 3×7 matrix with rank 3. Comment on the nature of solutions of A .

Question 5: Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ have matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \end{bmatrix}$$

Calculate the Rank, nullity and find a basis for the null space.

Question 6: For each statement below, determine whether it is true or false. Justify your answer with a clear explanation.

1. The vectors b that are not in the column space $C(A)$ form a subspace.

2. If $C(A)$ contains only the zero vector, then A is the zero matrix.
3. The column space of $2A$ equals the column space of A .
4. The column space of $A - I$ equals the column space of A .

Question 7: Consider the following system of linear equations:

$$\begin{aligned}x + 2y - z &= 3 \\2x - y + 3z &= 1 \\4x + 3y + z &= k\end{aligned}$$

1. Find the value(s) of the constant k for which this system is consistent (i.e., has at least one solution).
2. For the value of k found in part 1, what is the rank of the coefficient matrix and the ****rank**** of the augmented matrix? Use this to comment on the nature of the solution (unique or infinitely many).

Question 8: Let

$$v = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} \in \mathbb{R}^3,$$

and let $B = \{b_1, b_2, b_3\}$ be an orthonormal basis of $\mathbb{R}^3$ defined by

$$b_1 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad b_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad b_3 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \end{bmatrix}.$$

Find the coordinate vector $[v]_B$ of v with respect to the orthonormal basis B .

Question 9: Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$T(x, y, z) = (2x - y - z, 3x + y - 2z, x + 4y + z).$$

1. Find the matrix representation of T in the standard basis.
2. Compute the image of the standard basis vectors under T .
3. Verify that T satisfies the properties of a linear transformation.
4. Determine the kernel and image of T , and verify the rank-nullity relation.
5. Comment on whether T is invertible and justify your answer geometrically.

Question 10: Let W be the subspace of $\mathbb{R}^3$ spanned by the orthonormal vectors

$$u_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \quad \text{and} \quad u_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

1. Find the orthogonal projection of the vector $v = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$ onto the subspace W .
2. Find the distance between the vector v and the subspace W .